HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained production automation systems and associated interfaces for a pharmaceutical manufacturing site, ensuring compliance with regulatory standards.
- Executed system implementation and upgrade activities for **PAS, PLC, and BAS** projects, including software and hardware enhancements, and network improvements.
- Managed the lifecycle of automation systems, performing proactive troubleshooting and preventive maintenance to ensure operational continuity.
- Defined Standard Operating Procedures (SOPs) for automation services, enabling efficient and standardized operations across the engineering department.
- Collected and analyzed **Automation Department metrics/KPIs**, identifying areas for efficiency improvements and driving the adoption of new technologies.
- Collaborated with business relationship managers to ensure automation solutions effectively met evolving business requirements and strategic objectives.
- Containerized automation scripts and services using **Docker** for consistent deployment across development and production environments on **AWS EC2**.

SKILLS

Core Engineering – Automation Systems, PAS, PLC, BAS, System Implementation, Hardware Upgrades, Network Enhancements, Sensor Upgrades, Recipe Management, User Requirement Specification

IT & Infrastructure – IT Infrastructure & Hardware, Linux, Shell Scripting, Cron Jobs, AWS EC2, Docker

Methodologies & Standards – ISA S95, ISA S88, Industry 4.0, IIoT, Agile, Scrum, Change Management, Continuous Professional Development

Data & Analytics – Data Warehousing, Data Modelling, Business Intelligence Theory, Metrics, KPIs, Operational Reporting

Software Development – Java, Spring Boot, REST APIs, Python Scripting, SQL, PostgreSQL, MySQL, Oracle

Monitoring & Support – Production Support, Incident Response, SLA Management, Splunk, Dynatrace, CloudWatch

PROJECTS

Manufacturing Automation System Upgrade (HCL Technologies)

- Executed a phased upgrade of the primary **Process Automation System (PAS)** for a manufacturing client, involving **Siemens PLC** software updates and control logic adjustments.
- Developed Python scripts to automate data extraction and reporting from system logs, facilitating faster root cause analysis for production issues.
- Managed the change control process for system configuration updates, coordinating User Acceptance Testing (UAT) with operational teams.

Industrial IoT Data Pipeline for Predictive Maintenance (HCL Technologies)

- Designed and implemented a data pipeline using **Python** and **Pandas** to ingest sensor data from industrial equipment, preparing it for analysis.
- Developed SQL queries against a **PostgreSQL** database to validate data integrity and extract features for predictive maintenance models.
- Integrated data collection scripts with **Jenkins** for automated execution and monitoring of the data ingestion process.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)